INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

INVERTERS, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for inverters which convert electrical power in aircraft from nominal 28 Volts DC (see 6.4.2) power to three-phase or single-phase AC (see 6.4.1) power at 400 Hertz.

1.2 Classification. Inverters shall be of the following classes, as specified (see 3.7.2.1 and 6.2):

Class A. Inverters which are self-cooled and designed to meet the temperature-altitude requirements of Standard MS33543, curve II.

Class B. Inverters which are self-cooled and designed to meet the temperature-altitude requirements of Standard MS33543, curve I, except that the ambient temperature shall not exceed +85°C. The inverter shall deliver rated output up to 50,000 feet and 50 percent (minimum) rated power up to 65,000 feet.

Class C. Inverters which are blast-cooled and designed to meet the temperature-altitude requirements of Standard MS33542.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Electrical Systems Department (SY60), Patuxent River, MD 20670-5304, by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6125

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.3).

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-W-5088	Wiring, Aerospace Vehicle
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-T-7928	Terminal, Lug and Splice, Crimp Style Copper
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-15024	Plate Tags and Bands for Identification of Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-A-21180	Aluminum Alloy Castings, High Strength
MIL-C-39006/22	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85°C (Voltage Derated to 125°C), Established Reliability, Style CLR79
MIL-C-39006/25	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85°C (Voltage Derated to 125°C), Established Reliability, Style CLR81
MIL-R-39015	Resistors, Variable, Wirewound (Lead Screw Actuated) Established Reliability, General Specification for

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MIL-I-46058	Insulation Compound, Electrical (For Coating Printed Circuit Assemblies
MIL-E-81910	Electric Power and Control Equipment, Aircraft, Testing, General Specification for

STANDARDS

Federal

FED-STD-595	Colors Used in Government Procurement

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-198	Capacitors, Selection and Use of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Report, Preparation of
MIL-STD-970	Standards and Specifications, Order of Preference for the Selection of
MIL-STD-973	Configuration Management
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-1629	Procedures for Performing a Failure Mode Effects and Criticality Analysis
MIL-STD-2000	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-2073-1	DOD Materiel Procedures for Development and Application of Packaging Requirements
MIL-STD-2073-2	Packaging Requirement Codes
MIL-STD-2175	Castings, Classification and Inspection of

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MS3376	Nuts, Self-locking, Cres, 12 Point Captive Washer
MS33543	Criteria-Temperature and Altitude Range Self Cooled Electric Equipment
MS90415	Nut, Self Locking, Steel, 160 KSI, 450 DEG. F, 12 Point Captive Washer

(See Supplement 1 of this specification for list of associated MS sheets.)

MILITARY HANDBOOKS

MIL-HDBK-217	Reliability of Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND DOCUMENTS

NAVSEA TE000-AB-GTP-010	Parts Derating Requirements and Application Manual for Navy Electronic Equipment

(Copies are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated MS sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 MS sheets. The individual item requirements shall be as specified herein and in accordance with the applicable MS sheets. In the event of any conflict between requirements of this specification and the MS sheet, the latter shall govern. (If a specific requirement specified herein is not required for an inverter, it shall be so indicated on the MS sheet; for example "Shock - Not applicable.").

3.2 Classification of requirements. The requirements for inverters are classified herein as follows:

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Requirement	Paragraph
Qualification	3.3
Materials	3.4
Selection of parts and materials	3.5
Design and construction	3.6
Performance characteristics	3.7
Reliability	3.8
Markings	3.9
Workmanship	3.10

3.3 Qualification. Inverters furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4 and 6.3). Each unit delivered shall fully meet all the requirements of this specification and the applicable MS sheet, and shall be capable of successfully completing the tests required by this specification for quality conformance inspection and for inspections listed for any one of the qualification samples. The Naval Air Warfare Center Aircraft Division (SY60), Patuxent River, Maryland 20670-5304 is the qualifying activity for equipment submitted under this specification. Any design changes to the inverter shall be handled in accordance with MIL-STD-973, with copies of the documentation provided to the qualifying activity and the procuring activity.

3.4 Materials. Materials used in the manufacture of inverters shall be of high quality and suitable for the intended purpose.

3.4.1 Metals. Metals shall be of a corrosion resistant type or suitably processed to resist corrosion.

3.4.1.1 Dissimilar metals. Unless otherwise suitably protected, dissimilar metals such as brass, copper, or steel shall not be used in intimate contact with aluminum or its alloys. When protection is used, it shall be of such type that a low impedance path is offered to radio frequency currents. Dissimilar metals are defined as two metal specimens that generate an electric current in a conductive solution when in contact or otherwise electrically connected.

3.4.1.2 Magnesium alloy parts. Neither magnesium nor magnesium alloy parts shall be used.

3.4.1.3 Finishes and protective coatings.

3.4.1.3.1 Aluminum and aluminum alloy parts. Aluminum and aluminum alloy parts shall be anodically treated in accordance with MIL-A-8625 or chemically treated in accordance with MIL-C-5541. To provide adequate bonding, the aluminum oxide film formed by this treatment shall be removed from the areas under screws, nuts, etc. used for assembly or mounting purposes.

3.4.1.3.2 Steel parts. Steel parts shall be plated, if necessary, so that the inverter meets all of the specified electrical and environmental requirements.

3.4.1.3.3 Color. The inverter shall be finished in 37038 black in accordance with FED-STD-595.

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3.4.2 Non-metallic materials. Non-metallic materials used shall be non-toxic, fungus inert, flame resistant, and shall not support combustion.

3.5 Selection of parts and materials. Specifications and standards for all materials and parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the design and construction of inverters complying with the requirements of this specification, shall be selected in accordance with MIL-STD-970. All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance. The same part shall be used for as many applications as possible in a given unit and in units of different ratings to simplify the design.

3.5.1 Standard parts. Military standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their military part numbers. Any standard part, of the same part number, shall be readily interchangeable as applied within the inverter.

3.5.2 Non-standard parts. The use of non-standard parts is permitted in the event that there is no suitable standard part available on the date of invitation for bids, however, such use shall be approved by the qualifying activity. If a non-standard part is approved by the qualifying activity, subsequent approval is not necessary. Request for approval for the use of non-standard parts shall be submitted in writing and shall contain the following information:

a. Designation of equipment and assembly wherein part is used.

b. Reference designation of part (circuit, symbol, part number, as applicable).

c. Name and complete detailed technical description of part.

d. Part manufacturer's name and address.

e. Part manufacturer's drawing and part number.

f. Reason for not using standard part including, when applicable, a comparison of characteristics of the non-standard part with the standard part which most nearly possesses the required characteristics.

g. Other information required herein for specific types of parts.

3.5.3 Commercial (hardware) parts. Commercial utility hardware items: screws, bolts, nuts, and cotter pins may be used without approval provided they possess suitable properties and are replaceable by MS or AN parts without alteration, and provided the corresponding MS or AN part numbers are referenced in the parts list and on the manufacturer's drawings.

3.5.4 Electron tubes. Electron tubes of either the vacuum or gas filled type shall not be used.

3.5.5 Capacitors. Capacitors shall be selected and applied in accordance with MIL-STD-198. The use of wet slug tantalum capacitors, except for tantalum

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cased units in accordance with MIL-C-39006/22 and MIL-C-39006/25, requires the approval of the qualifying activity and, if applicable, the procuring activity. Silver cased tantalum capacitors shall not be used.

3.6 Design and construction. Dynamic protective devices (fuses, circuit breakers, and thermally actuated switches) shall not be used. Deviations from this specification or from the applicable MS sheet must be specifically approved by the procuring and qualifying activities. Any changes to the design of the inverter shall be handled as specified in 3.3.

3.6.1 Electrical.

3.6.1.1. Electric connections.

3.6.1.1.1 External electric connections. The equipment shall be so designed that external connections and wiring can be made in accordance with the wiring installation, ground return, and connectors requirements of MIL-W-5088. The equipment structure shall not be used as a current path except for radio noise shielding.

3.6.1.1.1.1 Connectors. When alternate connectors are specified, the inverter design shall be such that it may be loaded on either receptacle, but not simultaneously. This shall be accomplished by a simple, easily accessible means to electrically disconnect the unused receptacle, or by providing a suitable protective cover which will automatically cover the unused receptacle. The cover, if used, shall be designed so that it cannot be separated from the unit.

3.6.1.1.2 Internal electric connections. Internal connections shall be made in accordance with MIL-STD-454, Requirements 1, 10, 17, 19, 20, and 69. General purpose soldering shall be performed in accordance with MIL-STD-2000 Task A. For applications which include active surface mounted microelectronic devices, soldering shall be performed in accordance with MIL-STD-2000 Task F. For three-phase inverters, the design shall be such that no internal reconnections will be required to change from wye to delta. If input and output contactors are used, they shall be deenergized simultaneously.

3.6.1.2 Adjustments. No adjustment other than for voltage and frequency as provided for by 3.6.1.2.1 and 3.6.1.2.2 shall be required to put the unit into satisfactory operation at the time of the installation on the aircraft.

3.6.1.2.1 Voltage setting. When required by the applicable MS sheet, the inverter shall be constructed to permit the output voltage to be set over a range of at least ±5.0 percent of rated value in increments of not more than 0.5 percent of rated voltage. The setting means shall be so constructed that it will not be displaced by vibration or shock. The control shall be designed for adjustment using a straight screwdriver.

3.6.1.2.2 Frequency setting. No frequency adjustment capability will be required if the inverter has an internal control that automatically maintains the frequency over the life of the inverter. If a frequency adjustment capability is incorporated in the inverter, the control shall be constructed so that it will not be displaced by vibration or shock, and shall be adjustable with a straight screwdriver over the following ranges:

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ing type or suitably processed to withstand the environmental test requires-

(a) Inverters 250 VA and larger shall have an adjustable range at least ±2.5 percent of the rated frequency in increments of not more than 0.25 percent of the rated frequency.

(b) Inverters smaller than 250 VA shall have an adjustable range at least ±5.0 percent of the rated frequency in increments of not more than 0.25 percent of the rated frequency.

3.6.1.3 Input power contactor. An input power contactor, if used, shall not open at a voltage higher than 9 Volts DC at any temperature-altitude condition specified by the applicable MS sheet.

3.6.2 Mechanical.

3.6.2.1 Rotor. Rotating type inverters shall be so constructed that the rotor may be removed readily without special tools.

3.6.2.2 Brushes (brush type inverters only).

3.6.2.2.1 Brush accessibility. Brushes shall be accessible for replacement when brush opening covers are removed. Brush opening covers shall be easily accessible, and removable without special tools.

3.6.2.2.2 Brush fit. The face of each commutator brush shall contact its commutator 100 percent in the direction of rotation for at least 75 percent of brush dimension parallel to the shaft. The face of each slip ring brush shall contact its slip ring over an area that will insure satisfactory performance of the inverter; however, as a minimum requirement, slip ring brush contact area shall be 50 percent.

3.6.2.2.3 Cartridge type brushes. When cartridge type brush assemblies are used, there shall be an integral assembly of the brush, spring, and terminal which shall not rotate during installation or during service.

3.6.2.2.4 Brush position. Brush riggings for all units of the same part number by one manufacturer shall be completely interchangeable. Brush rigging in each inverter shall be pinned in that position which results in optimum commutation characteristics for that unit. The method of pinning shall not adversely affect interchangeability of parts between units, and shall allow the brush rigging position to be reset during overhaul.

3.6.2.2.5 Brush marking. Each brush shall be marked for allowable wear. Brushes having a cross sectional area parallel to the wearing surface larger than 1/16 square inch shall be marked for allowable wear by a readily discernible groove in its edge. Brushes having a cross sectional area parallel to the wearing surface of less than 1/16 square inch may be marked with a line. In either case, the groove or line shall be as shown in Figure 1.

3.6.2.2.6 Spare brushes. All spare brushes shall be formed to a radius equal to approximately 100 percent of the radius of the new commutator or slip ring, and shall be of the same specifications as the corresponding brushes approved by the qualification tests.

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3.6.2.2.7 Brush life. When used, brushes shall have a useful life of no less than 500 hours.

3.6.2.3 Shock mounts. Shock mounts or vibration isolators, if used, shall be of such design that the main support between the stationary and isolated member is of all metal construction.

3.6.2.4 Interchangeability. Provisions for interchangeability are required to the maximum extent possible. Design tolerances shall permit parts, subassemblies, and assemblies to be used in the inverter without regard to the source of supply or manufacturer. Parts, subassemblies, and assemblies having the full range of dimensions and characteristics permitted by the governing specification shall be usable as replacement items without selection and without departure from the specified performance requirements of the inverter. When permission is granted to use a non-standard part or material because the existing standard part or material is not available, the inverter shall be so designed that the non-standard part or material and the standard part or material are interchangeable. When the specification for the part or material contains substitutability or supersession information, the design shall permit the substitute or superseding parts or materials to be used interchangeably.

3.6.2.5 Screw threads. Screw threads shall conform to the requirements of MIL-S-7742.

3.6.2.6 Safety wiring and staking. Accidental loosening of screws, screw parts, and other connections shall be prevented by safety wiring (0.032 inch minimum OD) where practicable, staking, self-locking nuts, or other approved methods. Washers and cotter pins, where used, shall be assembled in a manner which prevents rotation of washers and movement of cotter pins under conditions of vibration.

3.6.2.7 Ventilation openings. All ventilation or other openings shall be designed to prevent passage of foreign objects, and shall be of such size to prevent passage of spherical objects of 5/16 inch diameter or larger.

3.6.2.8 Operating position. The inverter shall operate in any position.

3.6.2.9 Size and weight. The dimensions and weight shall be as specified in the applicable MS sheet. The weight and dimensions shown on the applicable MS sheet shall include the weights and dimensions of all auxiliary apparatus necessary to make the inverter conform to this specification and the applicable MS sheet. Where only maximum and minimum dimensions are shown, the part need not have the shape shown. But, the part, including all protrusions, shall be contained within the outline shown.

3.6.2.10 Lubrication. The inverter shall not require additional lubrication during 1000 hours of operation.

3.7 Performance characteristics. The inverter shall deliver power in accordance with the requirements specified herein and the applicable MS sheet. For three-phase units, the limits apply to each phase.

3.7.1 Electrical.

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3.7.1.1 Input power. The inverter shall meet the requirements of this specification when supplied with power as defined herein from an aircraft electrical system. The nominal input voltage is 28 Volts DC, or as specified on the applicable MS sheet.

a. Normal input voltage (all inverters) - The normal input voltage range is defined as the range encountered when all aircraft engines are operating and the aircraft electrical system is fully operational. The normal input voltage range is typically 26 to 29 Volts DC, and is specified in the applicable MS sheet.

b. Abnormal input voltage - The abnormal input voltage range, when applicable, is defined as the range encountered during operation from aircraft batteries or other emergency power sources. The abnormal input voltage range is typically 20 to 26 Volts DC, and is specified in the applicable MS sheet.

c. Low input voltage - The low input voltage range, when applicable, is defined as the range encountered during operation from aircraft batteries when those batteries are being used for engine starting. The low input voltage range is typically 9 to 18 Volts DC, and is specified in the applicable MS sheet.

3.7.1.2 Output voltage regulation. The inverter shall have a nominal output voltage of 115 Volts RMS, and shall maintain the steady state output voltages as indicated on the applicable MS sheet under all specified conditions.

3.7.1.3 Output frequency regulation. The inverter shall have a nominal frequency of 400 Hertz, and shall maintain the steady state frequency as indicated on the applicable MS sheet under all specified input voltages, load, temperature, and altitude conditions specified.

3.7.1.4 Efficiency. The efficiency of the inverter shall be within the limits specified on the applicable MS sheet.

3.7.1.5 Three-phase output.

3.7.1.5.1 Phase rotation. The phase rotation shall be in the order of A-B-C (see 4.7.2.4.2).

3.7.1.5.2 Output voltage unbalance. For all balanced and one-third unbalanced load conditions, the maximum output voltage unbalance shall be 3 Volts RMS. For two-thirds unbalanced load conditions, the maximum output voltage unbalance shall be 4 Volts RMS. Individual phase voltages (see 6.4.4) shall be within the limits specified in 3.7.1.2.

3.7.1.5.3 Phase displacement. The displacement between adjacent phases shall be 120 ± 4° for both balanced and unbalanced load conditions (see 4.7.2.4.3).

3.7.1.6 Recovery time. When rated full load (see 6.4.3) is applied and removed, the output voltage shall remain within the limits of Figure 2 (see 4.7.2.5).

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3.7.1.7 Input power application. For static inverters, the output voltage and frequency during input power application shall be within and remain within the limits of the applicable MS sheet within two seconds after application of input voltage. During the two second period, output voltage transients shall not exceed 140 VRMS. For determining compliance of voltage transients, the onset of the transient shall be considered as the time when the output voltage first reaches a value equal to minimum steady state limits. For rotary inverters that require time for the voltage and frequency controls to become effective, the uncontrolled voltage shall not exceed the limits shown in Figure 2, and the frequency shall not exceed 105 percent of the maximum controlled value specified by the applicable MS sheet after 0.25 seconds. During the 25 second warmup period of Figure 2, the output voltage shall not exceed the maximum steady state voltage as specified in the applicable MS sheet, except that one transient excursion of up to 2 seconds duration is permitted, not to exceed 140 VRMS. The onset of this transient shall be considered as the time when the output voltage first exceeds the maximum steady state voltage as specified in the applicable MS sheet. The input power application current for all inverters shall not exceed 1000 percent of full load current as computed from the minimum allowable efficiency specified on the applicable MS sheet or 1500 Amperes, whichever is less. The current shall return to and stay within 300 percent of full load value within 0.3 seconds after first application of input voltage (see 4.7.2.6).

3.7.1.8 Overload and short circuit capacity. Over the normal input voltage range (see 3.7.1.1a) the inverter shall be capable of operating without damage into any passive load impedance (including a short circuit) for five seconds. For three-phase units this shall apply to each phase separately, any two phases simultaneously, or all phases simultaneously. The inverter shall be capable of meeting all of the overload and short circuit requirements with two minutes separating the removal and subsequent application of an overload or short circuit (see 4.7.2.7).

3.7.1.8.1 Overload. Over the normal input voltage range, inverters rated 1000 VA or less shall, without damage, maintain the minimum output voltage specified by the applicable MS sheet while delivering 200 percent rated output current for five seconds. Inverters rated above 1000 VA shall maintain the minimum output voltage specified by the applicable MS sheet while delivering 150 percent rated output current for 15 seconds. For three-phase units, this shall apply to each phase separately, any two phases simultaneously, or all three phases simultaneously. During these conditions, no phase voltage shall exceed 117.5 Volts RMS.

3.7.1.8.2 Short circuit. Over the normal input voltage range, inverters rated 1000 VA or less shall be capable of delivering, without damage, at least 250 percent rated load current for five seconds. Inverters rated above 1000 VA shall be capable of delivering, without damage, at least 200 percent rated load current for five seconds. For three-phase units, short circuit requirements shall apply to one phase, two phase, and three phase line-to-line and line-to-neutral short circuits. During these conditions, no phase voltage shall exceed 117.5 Volts RMS.

3.7.1.8.3 Commutation (brush-type inverters only). Immediately following the runs of 3.7.1.8.1 and 3.7.1.8.2, with the unit hot, the commutation of all brushes shall be observed over the normal input voltage range for rated load,

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half load, and no load. There shall be no more than fine pinpoint sparking during this test.

3.7.1.9 Electromagnetic compatibility. When tested as specified in 4.7.2.8, the conducted and radiated electromagnetic interference shall be within the limits specified in MIL-E-81910.

3.7.1.10 Output modulation. (See 4.7.2.9)

3.7.1.10.1 Voltage. The output voltage modulation shall not exceed 1.0 percent under normal input voltage conditions at full load, half load, and no load.

3.7.1.10.2 Frequency. The inverter shall not develop sustained frequency oscillations under conditions of full load, half load, or no load.

3.7.1.11 Waveform. The output voltage crest factor shall not deviate from 1.414 ± 10 percent, nor the value of any harmonic observed exceed 5 percent of the fundamental.

3.7.1.12 Reduced input voltage operation. During the reduced input voltage test, the output voltage of inverters rated at 750 VA and below shall remain between 100 and 120 Volts RMS, and the frequency between 350 and 420 Hertz. For inverters rated above 750 VA, the output voltage shall remain between 100 and 120 Volts RMS, and the frequency between 370 and 410 Hertz (see 4.7.2.11).

3.7.1.13 Acoustical noise. When limits are specified by the applicable MS sheet and when tested as specified in 4.7.2.13, the maximum sound level measured at any point shall not exceed the specified limits.

3.7.1.14 Overspeed (rotary inverters only): The ability of the inverter to operate at 25 percent above rated speed shall be verified by test (see 4.7.2.13).

3.7.2 Environmental.

3.7.2.1 Temperature-altitude. The inverter shall be capable of operation without degradation of performance throughout the temperature-altitude range specified below for the class specified in the applicable MS sheet.

3.7.2.1.1 Class A. The inverter shall be capable of operation without degradation in specified performance throughout the temperature-altitude range of MS33543, Curve II.

3.7.2.1.2 Class B. The inverter shall be capable of operation without degradation in specified performance throughout the temperature-altitude range of MS33543, Curve I, except that no operation shall be required above +85°C. The inverter shall deliver rated output power up to 50,000 feet and 50 percent (minimum) rated output power to 65,000 feet.

3.7.2.1.3 Class C. The inverter shall be capable of operation without degradation in specified performance throughout the temperature-altitude range of MS33542.

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3.7.2.2 Acceleration. The output voltage and frequency of the inverter shall remain within the limits of 3.7.1.2 and 3.7.1.3 while operating during the acceleration test (see 4.7.2.14). There shall be no evidence of physical damage, ccccccccccccccccccccccccand the voltage and frequency regulation after the test shall be within the limits specified in 3.7.1.2 and 3.7.1.3.

3.7.2.3 Humidity. As a result of the humidity test there shall be no evidence of corrosion, peeling, pitting, blistering, or cracking of the finish. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3 (see 4.7.2.15).

3.7.2.4 Shock.

3.7.2.4.1 Basic design. The output voltage shall not exceed the limits of Figure 2 while operating during the basic design shock tests (see 4.7.2.16). There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3.

3.7.2.4.2 Crash safety. The inverter shall comply with the equipment crash safety mechanical shock requirements of MIL-E-81910.

3.7.2.5 Vibration. The output voltage modulation of the inverter measured during the test shall not exceed two percent (see 4.7.2.17). The output voltage and frequency measured during the test shall remain within the limits specified in 3.7.1.2 and 3.7.1.3. There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3.

3.7.2.6 Fungus. As a result of the fungus test the inverter shall show no evidence of fungus growth on any surface. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3 (see 4.7.2.18).

3.7.2.7 Salt fog. As a result of the salt fog test there shall be no evidence of corrosion, peeling, pitting, blistering, or cracking of the finish. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3 (see 4.7.2.19).

3.7.2.8 Dust. The inverter shall be capable of operating within the specified limits during the sand and dust test. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3 (see 4.7.2.20).

3.7.2.9 Explosive atmosphere. The inverter shall not cause ignition when tested as specified in 4.7.2.21.

3.8 Reliability. When operating under any temperature-altitude conditions specified by the applicable MS sheet, the inverter shall have a useful life of not less than 1000 hours. When used, brushes may be replaced after 500 hours of operation.

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3.8.1 Reliability development. The reliability of the inverter shall be demonstrated by test (see 4.7.3).

3.9 Markings.

3.9.1 Terminals. Markings of external electrical connections shall be as specified on the applicable MS sheet.

3.9.2 Rotation marking. On rotating-type inverters, the direction of rotation shall be plainly indicated by an arrow.

3.9.3 Wiring diagram. Attached to the inside of the control box cover or other surfaces, when practicable, there shall be a circuit diagram with a legend describing the electrical components of the equipment. This diagram shall include rated values of resistors and capacitors. The diagram shall be legible, protected against humidity, and not easily removed.

3.9.4 Airflow. When an integral cooling fan is used, direction of airflow shall be plainly and permanently marked on the outside of the housing. When the direction of the airflow is marked on the cover, provision shall be made to assure that the cover cannot be installed to indicate a false airflow direction.

3.9.5 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.9.6 Nameplate. A nameplate conforming to MIL-P-15024, Type A, B, C, G, or H, with the following markings, in addition to the requirements of MIL-STD-130, shall be securely attached to the inverter. The nameplate shall contain the following information:

Inverter	_____	VA		
Input:	_____	Volts DC	_____	Amperes DC
Output:	_____	Volts AC	_____	Amperes AC
	400	Hz	_____	Phase

Military Part No.
National Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Date of manufacture
Contract or Order No.
Manufacturer's name or trademark
Inspector's stamp
US

If the MS sheet provides for the separation of the inverter into a control component and a power component, each component shall be equipped with a nameplate bearing the above information plus the following:

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To be used with (control or power component)*
National Stock No.
Manufacturer's Part No.

*Manufacturer to insert applicable nomenclature

3.10 Workmanship. The inverter shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All inverters shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Qualification retention inspection (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 Qualification inspection.

4.4.1 Qualification tests. Qualification tests shall be conducted at a facility satisfactory to the activity responsible for qualification (see 6.3). Qualification tests shall consist of the examination, inspections, and tests specified in Table I, in the order shown or as otherwise approved by the qualifying activity.

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4.4.2 Qualification samples. The qualification test sample shall consist of at least four (4) units of each part number for which qualification is sought. The sample units shall be produced, to the maximum extent practicable, with equipments and procedures normally used in production, and shall be identified as specified (see 3.9). The samples shall have successfully completed the quality conformance inspections shown in Table II.

4.4.3 Data submission.

4.4.3.1 Drawings. Together with the test samples, the manufacturer shall submit two (2) reproducible copies of the outline and detail assembly drawings, schematics, and parts lists. Upon completion of qualification testing and prior to qualification approval, the manufacturer shall again provide two (2) reproducible copies of the outline and detail assembly drawings, schematics, and parts lists. Any changes in the reproducible copies from those submitted with the qualification samples shall be indicated in detail.

4.4.3.2 Certification. Together with the test samples, the manufacturer shall submit a materials inspection report consisting of a certification, supported by verifying data, stating that the materials and finishes specified in 3.4 and used in fabricating the inverters are in accordance with the applicable referenced specification or requirements prior to such fabrication. Also, the manufacturer shall submit documentation certifying that the inverter meets the explosive atmosphere requirements specified in 3.7.2.9.

4.4.3.3 Component stress level. Together with the test samples, the manufacturer shall submit a report listing all inverter electric/electronic components, their maximum rated stress, their applied stress within the inverter, and the ratio of applied to rated stress. An analysis of electric/electronic components shall be performed in accordance with MIL-HDBK-217 to demonstrate conformance to paragraph 3.8 of this specification. Data shall be based upon the most severe inverter environmental and operational conditions specified herein and the applicable MS sheet.

4.4.4 Disposition of qualification samples and test report. Upon completion of the qualification tests, the test samples shall be identified as required and forwarded to the qualifying activity (see 3.3 or 6.3). The test samples shall be accompanied by two (2) copies of the qualification test report, prepared in accordance with MIL-STD-831.

4.4.5 Retention of qualification. To retain qualification, the manufacturer shall comply with the requirements of 4.5. In the event that no production occurred during a twelve (12) month period, a report shall be submitted to the qualifying activity certifying that the company still has the capabilities and facilities necessary to produce the item. If during two (2) consecutive twelve (12) month periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product sample for testing in accordance with the qualification inspection requirements.

4.5 Qualification retention inspections. Each inverter of the same part number delivered as a qualified item under this specification shall be numbered sequentially in essentially the order that it is submitted for Government

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acceptance, regardless of the contract under which it is delivered or by whom it is purchased. To meet the delivery requirements for the first 50 or less inverters, the contract delivery requirement plus one additional inverter must successfully complete the required quality conformance inspections. The contract delivery of required inverters are to be shipped as directed by the applicable contract(s). The additional inverter (retention sample), selected at random by the Government Inspector, is to be forwarded to the Naval Air Warfare Center Aircraft Division, Electrical Systems Department (SY60), Patuxent River, MD 20670-5304 (Reference: 101). To meet the delivery requirements for each succeeding 150 inverters, 151 inverters must successfully complete the required quality conformance inspection, 150 of which are to be shipped as directed by the applicable contract(s). The additional inverter (retention sample) shall be randomly selected and forwarded as above. The span of serial numbers and the contract number of the production units from which each retention sample was selected shall be recorded, and this information included with the retention sample. Each retention sample shall successfully complete the quality conformance inspection shown in Table II, and the test data and test date shall also be included with the inverter (retention sample). The samples shall be shipped no later than the final unit in the production group from which they are selected. The units forwarded to the qualifying activity will be inspected and tested for conformance to this specification and the applicable detail specification. Tests will be conducted for each inverter (retention sample) according to any column from Table I at the discretion of the qualifying activity. All retention samples become Government property when accepted by the Government Inspector for shipment to the qualifying activity. The cost for each inverter (retention sample) shall be borne by the manufacturer. These requirements shall apply to all inverters procured in accordance with this specification whether procured directly by the Government or by private industry procuring for end item use by the Government.

4.5.1 Qualification retention samples. For each quotation and based on all production contracts obtained, it will be the manufacturer's responsibility to inform the procuring activity of the exact number of qualification retention samples required for forwarding to the qualifying activity. The number of qualification retention samples required should be taken into account when the contract bid is prepared. If the proper number of retention samples is not itemized in the contract, then the manufacturer will, without additional cost to the Government, forward to the qualifying activity whatever number of retention samples is determined by the cognizant Government Inspector to be required by paragraph 4.5 of this specification.

4.5.2 Rejection and retest of qualification and qualification retention samples. Samples which have been rejected or returned to the manufacturer for any reason during qualification or qualification retention tests, may be reworked or have parts replaced to correct defects. The manufacturer shall bear the cost of any repair or redesign work necessitated by a failure to meet the requirements. Before resubmitting the samples, full particulars concerning the rejection and the corrective action taken by the manufacturer must be submitted in writing by the manufacturer to the qualifying activity. Tests shall not be resumed until such a report is received. After the second attempt to meet one of the requirements of this specification or the applicable MS sheet, the manufacturer shall bear the cost of any additional retesting for that requirement. Where qualification tests are conducted under the auspices of the manufacturer, the qualifying activity will be advised upon failure of a qualification sample and

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the action taken by the manufacturer with regard to the failure. If the manufacturer does not correct and return to the qualifying activity inverters rejected during qualification or qualification retention testing within twenty-one (21) calendar days after the manufacturer receives them or does not submit a program acceptable to the qualifying activity for the correction of deficiencies noted within twenty-one (21) calendar days, the manufacturer will be notified of the proposed cancellation of the qualification program or proposed removal of the item from the Qualified Products List, as appropriate.

4.6 Quality conformance inspections. Quality conformance inspections shall consist of inspection of the product for delivery. Each inverter offered for delivery shall be subjected to the inspection specified in Table II. Data from these inspections shall be recorded and be made available for examination by authorized Government representatives.

4.6.1 Failure. Any inverter failing to comply with specified requirements during the quality conformance inspections shall not be delivered on the contract or purchase order. The manufacturer may take corrective action on the failed inverters as appropriate and resubmit for inspection. Reworked units shall be suitably marked to identify them as repaired items. In the event of failure during reinspection, further rework and resubmission shall not be allowed.

4.7 Methods of inspection.

4.7.1 Test conditions. Unless otherwise specified, tests shall be made under the following conditions.

4.7.1.1 Ambient temperature. Tests shall be performed at an ambient temperature of 25 ± 15°C.

4.7.1.2 Ambient altitude. Tests shall be run at approximately sea level altitude.

4.7.1.3 Tolerance and accuracy. The tolerance of test conditions, other than room ambient, and the accuracy of test apparatus shall be as specified in the test facilities requirements of MIL-E-81910.

4.7.1.4 Ambient temperature measurement. When tests are required to be conducted at reduced or elevated temperatures, the ambient temperature shall be determined by measurements of the air at several points surrounding, but not above, the inverter, and through averaging the results. For units without fans, air temperature measurements shall be taken 1 ± 1/4 inch from the surface of the inverter. For units with an internal fan, air temperature measurements taken 1 ± 1/4 inch from the fan inlet shall be considered the ambient temperature.

4.7.1.5 Airflow. For all maximum temperature-altitude tests below 20,000 feet conducted on inverters without integral cooling fans, the airflow velocity over the inverter within the test chamber shall be adjusted to approximate natural convection conditions and shall not exceed 100 feet/minute, but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance. For tests above 20,000 feet altitude, the airflow may exceed 100 feet/minute, but shall be adjusted to the minimum velocity

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consistent with maintaining the required ambient temperature within specified tolerance.

4.7.1.6 Stabilization. Stabilization of test conditions based on constant input voltage and ambient temperature is defined to have occurred when the change over a ten (10) minute period is measured to be less than:

a. Output voltage: 0.04 Volts RMS

b. Frequency: 0.1 Hz

c. Inverter critical component temperature: 0.5°C

4.7.1.7 Voltage measurement. The input and output voltage shall be measured at the terminals of the unit.

4.7.1.8 Position. The units shall be mounted base down.

4.7.1.9 Loading. Prior to testing, the inverter shall be operated with nominal input voltage at full rated load (see 6.4.3) until stabilized. The test shall be performed with the inverter operating at full rated load.

4.7.1.9.1 Load power factor. Unity power factor load shall be applied.

4.7.1.9.2 Multiple phase loading. For three-phase tests, the loads on each of the three phases shall be balanced with each phase current within 3 percent of the average magnitude of all three phase currents.

4.7.1.9.3 Definition of load. Single-phase load shall be considered the product of the RMS line current and the RMS voltage measured at the terminals of the inverter. Three-phase load shall be the $\sqrt{3}$ times the product of the average RMS line current and the average RMS line-to-line voltage measured at the terminals of the inverter.

4.7.1.9.4 Definition of full load. Full load shall be the maximum continuous load specified at the applicable input voltage and temperature-altitude conditions specified in the applicable MS sheet.

4.7.1.10 Input voltage. Nominal rated input voltage shall be applied (see 3.7.1.1).

4.7.1.11 Adjustment of frequency and voltage. No adjustment of frequency or voltage shall be made during any test.

4.7.2 Methods of test.

4.7.2.1 Visual and mechanical inspection. The inverter shall be inspected to determine compliance with the requirements of this specification and the applicable MS sheet with respect to materials, design and construction, parts, workmanship, dimensions, weight, and markings for which there are no applicable tests.

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4.7.2.2 Voltage and frequency regulation. The output voltage and frequency shall be measured under the conditions specified in 4.7.2.2.1 through 4.7.2.2.6. For each load, input voltage, position, and temperature-altitude condition, output voltage and frequency shall remain within the limits specified (see 3.7.1.2 and 3.7.1.3). However, in order to expedite testing inverters having a voltage adjustment, the voltage may be as much as 5 percent outside of the limits specified on the applicable MS sheet, provided that the total deviation during test does not exceed the difference between the specified limits. The frequency for inverters having a frequency adjustment may be outside of the limits specified on the applicable MS sheet by as much as 5 percent, provided that the total deviation during test does not exceed the difference between the specified limits. For inverters with brushes, there shall be no more than fine pinpoint sparking at the brushes during this test. When the applicable MS sheet places limits on the Freq/V ratio rather than on frequency and voltage separately, then the results of the above tests will be interpreted accordingly and the specified limits must be met. During the course of the tests, no components of the inverter shall exceed their maximum rated temperatures. Tests may be conducted at any intermediate input voltage, load, or temperature-altitude condition within the ranges of MS33543 Curve I or II or MS33542 (as applicable) at the option of the activity responsible for qualification.

4.7.2.2.1 Voltage and frequency regulation at room ambient temperature. The following tests, (a) through (c) below, shall be conducted immediately upon energizing the inverter, and then repeated after operating the inverter at nominal input voltage, full load, unity power factor until the output voltage and frequency are completely stabilized.

a. At the maximum normal input voltage (see 3.7.1.1a), the load shall be varied from no load to rated full load in steps of one-fourth rating. At each load condition, the inverter shall demonstrate its ability to activate.

b. At nominal rated input voltage (see 3.7.1.1), the test specified in 4.7.2.2.1a shall be repeated.

c. At the minimum normal input voltage (see 3.7.1.1a), the test specified in 4.7.2.2.1a shall be repeated.

4.7.2.2.2 Power factors. The tests specified in 4.7.2.2.1 shall be repeated at the minimum lagging and minimum leading power factors specified in the applicable MS sheet. Tests will be made at any other load power factors between minimum lagging and minimum leading at the option of the activity responsible for qualification.

4.7.2.2.3 Voltage and frequency regulation at maximum temperature-altitude conditions. The inverter shall be tested under each of the temperature altitude conditions listed in Table III. Prior to being energized, the inverter shall be soaked at the initial maximum temperature, sea level condition for at least one-half hour. Immediately after energizing the inverter, the test specified in 4.7.2.2.1 shall be conducted. The tests of 4.7.2.2.1 and 4.7.2.2.2 shall then be conducted after operating the inverter at nominal input voltage, full load, unity power factor until inverter output voltage, frequency, and temperature are completely stabilized. For each subsequent temperature-altitude condition, the test of 4.7.2.2.1 need only be conducted after inverter stabilization.

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4.7.2.2.4 Voltage and frequency regulation at minimum temperature-altitude conditions. The tests specified in 4.7.2.2.3 shall be repeated substituting the temperature-altitude conditions of Table IV and increasing the soak time at the initial -55°C sea level condition to two hours.

4.7.2.2.5 Operating position. The inverter shall be operated at nominal input voltage, full load, unity power factor in the normal base down position until output voltage, frequency, and critical component temperatures stabilize. Output voltage, frequency, and critical component temperature shall be measured as the inverter is rotated by increments of 90 degrees from the normal base down position, 360 degrees about each major axis except the vertical axis. Each position shall be held until output voltage, frequency, and critical component temperatures stabilize.

4.7.2.2.6 Worst condition. The inverter shall be operated at the worst indicated conditions determined from the tests specified in 4.7.2.2.1 through 4.7.2.2.5 over the normal input voltage range. The conditions from each of these tests resulting in highest output voltage shall be simultaneously applied. Prior to the test, the inverter shall be soaked at the applicable ambient temperature for a minimum of two hours prior to energizing the inverter. The inverter shall be operated until voltage, frequency, and critical component temperatures stabilize. The above test shall be repeated for minimum voltage, maximum frequency, minimum frequency, and maximum critical component temperature conditions.

4.7.2.3 Efficiency. The power output and input of the unit shall be measured at full load, unity power factor, sea level, and nominal input voltage. The efficiency shall be calculated under these conditions and shall not be less than specified in 3.7.1.4 under stabilized test conditions as defined in 4.7.1.6. When computing efficiency, mean (average) values of input voltage and current shall be used, and root mean square (RMS) values of the output voltage and current shall be used (see 3.7.1.4).

4.7.2.4 Three-phase output (three-phase inverters only). (see 3.7.1.5).

4.7.2.4.1 Output voltage unbalance.

4.7.2.4.1.1 Definition. Output voltage unbalance is defined as the maximum difference between phase voltage magnitudes.

4.7.2.4.1.2 Unbalanced loads. The effects of single-phase and unbalanced three-phase loads on the balance of the three-phase voltage output of the unit shall be determined as follows at minimum, nominal, and maximum normal input voltage.

a. When the unit is carrying no three-phase load, a single-phase line-to-neutral unity power factor load requiring 1/3 and then 2/3 of rated full load output current (a load of approximately 1/9 and 2/9 of the three-phase rating of the unit) shall be connected successively to each phase.

b. When the unit is carrying a unity power factor balanced three-phase load requiring 1/3 of the rated full load output current, a single-phase line-to-neutral unity power factor load requiring 1/3 and then 2/3 of the rated full load

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output current (a load of approximately 1/9 and 2/9 of the full three-phase rating of the unit) shall be connected successively to each phase.

c. When the unit is carrying a unity power factor balanced three-phase load requiring 2/3 of the rated full load output current, a single-phase line-to-neutral unity power factor load requiring 1/3 of the rated full load output current (a load of approximately 1/9 of the full three-phase rating of the unit) shall be connected successively to each phase.

The maximum value of the output voltage unbalance and the phase voltage occurring during parts (a) through (c) above shall not exceed the limits specified in 3.7.1.5.2.

4.7.2.4.1.3 Balanced loads. For all balanced loads within the power factor and load rating of the unit, the three-phase output voltage shall comply with 3.7.1.2 and 3.7.1.5.2. This shall be demonstrated during the test specified in 4.7.2.2.

4.7.2.4.2 Phase rotation. In conjunction with the test specified in 4.7.2.4.1, the phase rotation shall be determined and verified for compliance with 3.7.1.5.1.

4.7.2.4.3 Phase displacement angle. In conjunction with the tests specified in 4.7.2.2.1, 4.7.2.2.2, and 4.7.2.4.1.2, the phase displacement angle shall be determined and verified for compliance with 3.7.1.5.3.

4.7.2.5 Recovery time. The output voltage transient shall be measured when rated full load is applied and removed at unity and minimum lagging power factors. The unit shall not develop sustained oscillations and the output voltage shall remain within the limits specified in 3.7.1.6. This test shall be conducted at minimum, room, and maximum ambient temperatures specified at approximately sea level pressure and minimum, nominal, and maximum normal input voltages specified in 3.7.1.1a (see 3.7.1.6).

4.7.2.6 Input power application (see 3.7.1.7).

4.7.2.6.1 Minimum voltage. By the addition of external resistance in the line, the voltage input to the inverter from a regulated 28 VDC source shall be reduced to the minimum abnormal input voltage (see 3.7.1.1b). The inverter shall demonstrate the ability to activate and operate continuously with this external resistance at zero, half, and full load.

4.7.2.6.2 Input power application current and voltage. The input power application current and output voltage shall be measured at minimum, nominal, and maximum normal input voltage (see 3.7.1.1a) while initially applying power to the unit at room ambient temperature at zero, half, and full rated load at unity and minimum lagging power factors. The input power application current and voltage measurements shall indicate compliance with the limits specified in 3.7.1.7. This test shall be conducted at minimum, maximum, and room ambient temperatures specified at approximately sea level pressure.

4.7.2.6.2.1 Cyclic operation. The inverter shall be subjected to 50 consecutive 5-minute cycles of 1 minute operation, and 4 minutes inoperative. The

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Function	Monitor trippoint (disconnect external power

voltage and frequency shall be within the limits specified in 3.7.1.2 and 3.7.1.3 within 1 second after each energizing operation, except the first. Upon completion of this test, the inverter shall pass the tests specified in 4.7.2.2.1a, b, and c. No parts shall show signs of excessive burning or wear.

4.7.2.7 Overload and short circuit capacity. The inverter shall be tested to demonstrate compliance with the overload and short circuit capacity requirements specified in 3.7.1.8. Overload tests shall be conducted at both unity and minimum lagging power factors. The five second (or 15 second as applicable) overload or short circuit test shall be conducted at maximum, minimum, and room ambient temperatures specified at approximately sea level pressure (see 3.7.1.8).

4.7.2.8 Electromagnetic compatibility. Conducted and radiated electromagnetic interference shall be measured in accordance with the electromagnetic interference test of MIL-E-81910. During this test, the inverter load power factor shall be adjusted such that the combined power factor of the load and all auxiliary test equipment required shall present a total load of unity power factor to the inverter. The test shall be conducted at full and half load, at minimum and maximum normal input voltage (see 3.7.1.9).

4.7.2.9 Output modulation. The unit shall be operated at the conditions specified in 4.7.2.2.1 through 4.7.2.2.4 and shall meet the requirements of 3.7.1.10, except that frequency modulation need only be measured during the tests of 4.7.2.2.1 and 4.7.2.2.2 (see 3.7.1.10).

4.7.2.9.1 Voltage modulation. Total output voltage modulation shall be determined by means of a calibrated oscilloscope or other means satisfactory to the responsible qualifying activity. The percent modulation shall be computed as follows:

$$\text{Percent modulation} = \frac{E_{max} - E_{min}}{E_{max} + E_{min}} \times 100\%$$

Emax and Emin are the maximum and minimum peak voltages respectively, measured for the same polarity during a one second time period.

4.7.2.9.2 Frequency modulation. Output frequency modulation shall be determined by a method acceptable to the responsible qualifying activity. The percent modulation shall be computed as follows:

$$\text{Percent modulation} = \frac{F_{max} - F_{min}}{F_{max} + F_{min}} \times 100\%$$

Fmax and Fmin are the maximum and minimum instantaneous frequencies, respectively, attained during a 15 second time period.

4.7.2.10 Waveform (see 3.7.1.11).

4.7.2.10.1 Distortion. The harmonic and extraneous frequency content of the output voltage, per phase for three-phase units and across the two output lines for single-phase units, shall be determined under the conditions specified below. Compliance to the requirements of 3.7.1.11.1 and 3.7.1.11.2 shall be determined.

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a. Under the test conditions specified in 4.7.2.2.1a, b, and c with unity power factor, steady-state load.

b. Under the test conditions specified in 4.7.2.2.1a, b, and c with steady-state minimum lagging power factor load specified in the applicable MS sheet.

c. Tests (a) and (b) above, shall be repeated at maximum and minimum rated temperatures at approximately sea level pressure.

4.7.2.10.2 Crest factor. The crest factor shall be measured under the test conditions specified in 4.7.2.2.1 and 4.7.2.2.2 to verify compliance with 3.7.1.11. The crest factor limits apply to either polarity separately.

4.7.2.11 Reduced input voltage operation. The inverter shall demonstrate its ability to meet the requirements of 3.7.1.12. Inverters rated 750 VA and below, with the exception of 10 VA inverters, shall demonstrate their ability to start and to operate continuously under full load at input voltages from 20 to 29 Volts DC. Inverters rated above 750 VA shall start and operate with no load for at least 5 seconds with 22 Volts DC input voltage (see 3.7.1.12).

4.7.2.12 Acoustical noise. The acoustical noise of the inverter shall be measured in planes one foot from each of the six sides. In each plane, measurements shall be made opposite the center of the inverter surface to determine the sound level. The maximum sound level at all points shall be measured using the B weighted scale of an integrating-type sound level meter. Microphone orientation shall be in accordance with the recommendations of the manufacturer of the sound level meter to obtain the required frequency response. The calibrating devices or standards used shall be directly traceable to the National Institute of Science and Technology. Measurements shall be made under anechoic conditions with the inverter supported in a manner to minimize acoustic transmission and reflection by the support structure. Compliance to 3.7.1.13 shall be determined (see 3.7.1.13).

4.7.2.13 Overspeed (rotary inverters only). The inverter shall be operated at 25 percent above rated speed at no load for five minutes without mechanical failure. Immediately following this test, the inverter shall pass the tests specified in 4.7.2.2.1a, b, and c (see 3.7.1.14).

4.7.2.14 Acceleration. The inverter shall be tested in accordance with the acceleration test of MIL-E-81910. The inverter shall be operated at 28 VDC input and full load during Procedure II, and the output voltage and frequency shall be monitored during the test to determine compliance with 3.7.2.2. At the conclusion of the test, the inverter shall be examined for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c (see 3.7.2.2).

4.7.2.15 Humidity. The inverter shall be tested in accordance with the humidity test of MIL-E-81910. Following the test, the voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c. The inverter shall be inspected for evidence of damage (see 3.7.2.3).

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4.7.2.16 Shock. The inverter shall be tested in accordance with the mechanical shock test of MIL-E-81910, except that the inverter shall be operating during the basic design test at full load and 28 VDC input. The output voltage shall be monitored during the test to determine compliance with 3.7.2.4.1. At the conclusion of the test, the inverter shall be inspected for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c. The inverter shall not be operated during the crash safety test (see 3.7.2.4).

4.7.2.17 Vibration. (see 3.7.2.5).

4.7.2.17.1 Inverters weighing less than 35 lbs. The inverter shall be subjected to durability and performance tests as follows:

4.7.2.17.1.1 Durability. The inverter shall be subjected to sinusoidal vibration applied to the mounting base. The amplitude shall be .015 inch (maximum total excursion .03 inch). The frequency of vibration shall be established by cyclic operation from approximately 10 Hertz to 55 Hertz and back to 10 Hertz. The duration of one such cyclic operation shall be between 1 and 5 minutes. The inverter shall be vibrated continuously for 2 hours in the direction of each of the three major axes. At the end of the test period, the inverter shall be thoroughly inspected for damage or other defects resulting from the vibration test. Following the test, the inverter shall pass the tests specified in 4.7.2.2.1a, b, and c.

4.7.2.17.1.2 Performance. The inverter shall be subjected to sinusoidal vibration applied to the mounting base. The amplitude shall be .015 inch (maximum total excursion .03 inch) up to 60 Hertz. Above 60 Hertz the acceleration shall be limited to 10g. The inverter shall be vibrated for one excursion over the frequency range of 20 to 500 Hertz along each of its major axes. The vibration excursion shall take 12 to 15 minutes for each axis with no one frequency held for more than 3 minutes. The output voltage modulation shall not exceed two percent, and output voltage and frequency shall not be outside the limits of 3.7.1.2 and 3.7.1.3 at any time during the test. Following the test the inverter shall pass the tests specified in 4.7.2.2.1a, b, and c, and shall be thoroughly inspected for cracking or rupture of any components, solder connections, or of the mounting assembly itself. The presence of such damage shall be cause for rejection.

4.7.2.17.2 Inverters weighing 35 lbs or more. The inverter shall be subjected to the following tests. The maximum acceleration over the entire vibration envelope shall not exceed 5 g's during the following tests.

4.7.2.17.2.1 Resonance. The inverter shall be operated at full load during this test. Resonant frequencies of the inverter shall be determined by varying the frequency of applied vibration slowly through the specified range of frequencies at vibratory accelerations not exceeding those shown in Figure 3. Individual resonant frequency surveys shall be conducted with vibration applied along each of any set of three mutually perpendicular axes of the inverter. The inverter shall be vibrated at the indicated resonant conditions for the periods shown in the vibration test schedule of Table V, and with the applied double amplitude or vibratory acceleration specified in Figure 3. These periods of vibration shall be accomplished with vibration applied along each of three

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mutually perpendicular axes of vibration. When more than one resonance is encountered with vibration along any one axis, each resonance shall be sustained for the period shown in the applicable portion of the vibration test schedule. If more than four resonances are encountered with vibration applied along any one axis, the four most severe resonances shall be chosen for test. Cycling time indicated in Table V shall include the time required to perform the resonant frequency survey. Following the test, the unit shall pass the tests specified in 4.7.2.2.1a, b, and c.

4.7.2.17.2.2 Cycling. A vibration test shall be conducted with the frequency cycling between 10 and 500 Hertz in 15 minute cycles at an applied double amplitude of 0.036 inches or an applied acceleration of ±5g, whichever is the limiting value. Vibration shall be applied along each of the three mutually perpendicular axes according to the vibration test schedule of Table V. Following the test, the unit shall pass the tests specified in 4.7.2.2.1a, b, and c.

TABLE V. Vibration test schedule.
(Times shown refer to one axis of vibration)

Number of Resonances	0	1	2	3	4
Total vibration time at resonance*	-	½ hr	1 hr	1 ½ hr	2 hr
Cycling time	3 hr	2 ½ hr	2 hr	1 ½ hr	1 hr

*½ hr at each resonance.

4.7.2.18 Fungus. The inverter shall be tested in accordance with the fungus test of MIL-E-81910. At the conclusion of the test, the inverter shall be inspected for fungus growth, and the output voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c (see 3.7.2.6).

4.7.2.19 Salt fog. The inverter shall be tested in accordance with the salt fog test of MIL-E-81910. Specified drying shall be allowing the unit to stand at room ambient conditions for 24 hours. Output voltage and frequency regulation shall then be measured as specified in 4.7.2.2.1a, b, and c. This regulation test shall be repeated after 48 hours additional exposure at room ambient conditions (see 3.7.2.7).

4.7.2.20 Dust. The inverter shall be tested in accordance with the dust test of MIL-E-81910. The inverter shall be operated at no load with nominal input voltage (see 3.7.1.1), when required. At the conclusion of the test, the output voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c. For units equipped with a cooling fan, proper operation (speed and noise) shall be verified (see 3.7.2.8).

4.7.2.21 Explosive atmosphere (when specified). The inverter shall be tested in accordance with MIL-STD-810, Method 511.3, Procedure I (see 3.7.2.9).

4.7.3 Reliability development. The reliability development test shall consist of the tests specified in 4.7.3.1, 4.7.3.2, and 4.7.3.3. The test shall be structured as a test, analyze and fix (TAAF) program wherein any failure (any

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operation outside of specified limits) which occurs is analyzed and corrective action implemented by the manufacturer to prevent recurrence of the failure (see 3.8).

4.7.3.1 Endurance. The inverter shall be operated for 200 hours, during which time full rated load shall be switched alternately on and off for 5 second periods. During this test, readings shall be taken at no more than 12 hour intervals to demonstrate that the output voltage and frequency do not drift more than 2 percent during the entire test or any observed interval thereof. Following this test, the inverter shall pass the tests specified in 4.7.2.2.1a, b, and c, and 4.7.2.5. No part of the inverter shall show excessive wear.

4.7.3.2 Performance. The inverter shall be operated under the following conditions at full rated load unless otherwise indicated. New brushes, when used, may be installed for this test. The inverter shall pass the voltage and frequency regulation test specified in 4.7.2.2.1a, b, and c after the completion of each cycle. This test shall consist of four continuous cycles conforming to the following schedule with the maximum load, temperature, and altitude as specified by the applicable MS sheet:

6 hours at sea level

6 hours at 20,000 feet

6 hours at 35,000 feet

6 hours at 50,000 feet

6 hours at maximum altitude

The inverter shall be inspected for signs of overheating, excessive deterioration, and wear. The total wear for all components, except brushes, shall not exceed 12 percent of the allowable wear for the four cycles. There shall be no evidence of grooving or other surface damage to the face of the brush. Failure during this test shall be cause for rejection.

4.7.3.3 Life. The inverter shall be operated for 1000 hours. For the first 25 hours, it shall deliver rated current while operating at the maximum ambient temperatures(s) indicated by the applicable MS sheet. For the remainder of the test, the unit shall operate in an ambient temperature of 25°C ± 15°C. Inverters having a rated output of less than 100 VA shall operate with the mounting base in a position other than directly below the inverter during 500 hours of this test. Data shall be recorded and the inverter shall be brought to a complete stop and started under full load within three minutes at least every 25 hours during the test to demonstrate that the output voltage and frequency do not drift more than two percent during any 100 hour period, and no more than four percent during the entire test or any observed interval thereof. During the last hour of the first 25 hour period and following this test, the inverter shall pass the tests specified in 4.7.2.2.1a, b, and c. For this regulation test, the voltage and frequency may be adjusted if adjusting means are provided externally to compensate for the life test. No lubrication or maintenance of any sort, other than that permitted by 3.7.1.2, 3.7.1.3, and 3.6.2.2.7 shall be allowed during this test.

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4.7.4 Disassembly and inspection. At the conclusion of the test program, each sample shall be disassembled, as necessary, to inspect for evidence of wear, defects, and overheating. The presence of such evidence shall be cause for rejection.

5. PACKAGING

5.1 Preservation and packaging. The inverter shall be packaged in accordance with MIL-STD-2073-1, level A or C as specified in the contract order, unless directed otherwise by the procurement activity (see 6.2).

5.1.1 Level A. For level A packaging, preservation shall be in accordance with MIL-P-116, Method II, without preservation compounds, using metal reusable containers unless otherwise specified.

5.1.2 Level C. For level C the inverter shall be preserved and packaged individually in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be in accordance with MIL-STD-2073-1 ,level A, B,or C as specified in the contract or order, unless directed otherwise by the procurement activity (see 6.2).

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-2073-2 and MIL-STD-129, unless directed otherwise by the procurement activity (see 6.2).

5.3.1 Reinspection dates. Reinspection date marking shall be as specified by the procuring activity.

6. NOTES AND CONCLUDING MATERIAL

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The items described herein are intended for use on military aircraft to convert 28 Volt DC power to either single or three-phase, 115 VRMS, 400 Hertz AC power. Inverters typically are used to provide AC power on aircraft with main electrical generating systems that provide DC power.

6.2 Acquisition requirements.

a. Title, number, and date of this specification.

b. Class.

c. Military part number.

d. Required level of packaging and packing.

e. Type of container.

f. Reinspection date marking and any special marking.

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g. Handbooks or manuals, when required.

h. That the inverter shall have a minimum time to first maintenance action of 200 flight hours.

i. That the manufacturer shall include the cost of units required for retention of qualification (see 4.5) in the amount of his bid.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for the opening of bids, qualified for inclusion in the applicable Qualified Products List - whether or not such products have actually been so listed by that date. The attention of the manufacturers is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command (AIR-5362), Department of the Navy, 1421 Jefferson Davis Highway, Arlington, VA 22243-0700; however, information pertaining to qualification of products may be obtained from the qualifying activity - the Naval Air Warfare Center Aircraft Division, Electrical Systems Department (SY60), Patuxent River, MD 20670-5304.

6.3.1 Qualification retention samples. For each quotation and based on all production contracts obtained, it will be the manufacturer's responsibility to inform the procuring activity of the exact number of qualification retention samples required for forwarding to the qualifying activity (see 4.5). The number of qualification retention samples required should be taken into account when the contract bid is prepared. If the proper number of retention samples is not itemized in the contract, then the manufacturer will, without additional cost to the Government, forward to the qualifying activity whatever number of retention samples is determined by the cognizant Government Inspector to be required by paragraph 4.5 of this specification.

6.4 Definitions.

6.4.1 Alternating voltage and current. Alternating voltages and currents used in this specification shall be root mean square (RMS) values, unless otherwise specified.

6.4.2 Direct voltage and current. Direct voltages and currents used in this specification shall be mean (average) values.

6.4.3 Full rated load (full load). Full rated load used in this specification shall be the maximum continuous inverter output load current specified in the applicable MS sheet for a given temperature-altitude condition.

6.4.4 Phase voltage. Phase voltage shall be considered to be line-to-neutral for wye connection and line-to-line for delta connection.

6.5 Installation. The aircraft manufacturer shall allow clearance of at least one inch at each end of the inverter in addition to the maximum dimensions shown on the applicable MS sheet for installation and ventilation.

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6.6 Three-phase application. For three-phase aircraft application, the load on any one phase shall not exceed one third of the total three-phase inverter rating on a steady-state basis.

6.7 Subject term (key word) listing.

AC supply
Electrical
Rotary inverter
Single-phase
Static inverter
Static supply
Three-phase

6.8 Design guidelines.

6.8.1 Castings. Castings should be in accordance with MIL-STD-2175 and should be Class 3, Grade C or better. Aluminum alloy castings should be in accordance with MIL-A-21180. Defects not materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnating, welding, or other methods acceptable to the procuring activity. Inspection of repaired castings should be governed by quality control techniques subject to the approval of the procuring activity.

6.8.2 Semiconductor devices. Military standard semiconductor devices should be chosen, applied, and reported in accordance with Requirement 30 of MIL-STD-454 and should be selected from those meeting MIL-S-19500 and listed in QPL-19500. The reliability level of devices selected should be in the following order of preference beginning with the most preferred: JANTXV, JANTX, JAN. Requests for approval to use non-standard semiconductors should be accompanied by substantiating reliability data or proposed testing to assure adequate device reliability for the application.

6.8.3 Microelectronic devices. Microelectronic devices should be chosen, applied, and reported in accordance with Requirement 64 of MIL-STD-454 and should be listed in MIL-STD-1562. Requests for approval to use devices which do not comply with this requirement should be submitted as non-standard parts, and should be accompanied by substantiating reliability data or proposed testing to assure adequate device reliability for the application.

6.8.4 Potentiometers and adjustable resistors. Only sealed units conforming to MIL-R-39015 should be used.

6.8.5 Terminal block design. Where terminals are used for the external connection to inverters, they should be of the stud type, and should be designed so that the current is conducted by means of surface-to-surface contact, and not through the stud threads. All studs should be steel, corrosion resistant, C-34 Rockwell minimum hardness, and should be of sufficient length to accommodate two MIL-T-7928 lugs. Nuts used on terminals should conform to either MS3376 or MS90415. There should be no dielectric material in the compression build-up. The terminal block should be so designed that it can be removed and replaced on the inverter without the necessity of rebrazing or soldering. Barriers affording a positive separation of leads and terminals should be provided on the terminal

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block. Durable, reusable, non-conductive terminal covers should be provided. Terminal designations should be durably, legibly, and prominently marked on the terminal block.

6.8.6 Maintainability. The inverter should be designed to facilitate assembly, disassembly, location of trouble sources, and maintenance without aid of special tools. All functional parts should be readily identifiable, accessible, and removable for replacement. When printed circuit boards are utilized, only clear transparent conformal coating conforming to MIL-I-46058 type UR or XY should be used in order to permit identification, location, removal, and replacement of functional parts. Functional parts are defined as individual component parts such as resistors, capacitors, semiconductor devices, inductive devices, or replaceable functional assemblies such as active or passive filters, voltage dividers, dual transistors or diodes.

6.8.7 Failure mode and effects analysis (FMEA). A failure mode and effects analysis should be performed in accordance with the requirements of MIL-STD-1629 Task 101. Inverter protective functions should be independent of the control functions, and no single fault or failure should result in inverter operation outside the protective limits specified herein.

6.8.8 Cooling. The inverter should be self-cooled. No conduction cooling to the inverter environment should be allowed. Safe operating equilibrium temperature should be maintained for the inverter over the specified range of operating conditions, and the rated temperature of individual inverter components should not be exceeded. The use of an internally mounted, integral cooling fan is permitted for inverters with ratings of 750 VA and greater. The fan should use a motor having ball bearings of quality level ABEC-3 or higher, and the fan should be rated for operation at 100°C ambient or higher.

6.8.9 Reliability. Reliability should be considered in each phase of the inverter design process. Techniques used to determine and evaluate the detailed design should include a stress analysis with associated predictions and allocations in conformance with MIL-HDBK-217; sneak circuit and worst case analysis on all inverter control and protective circuits in accordance with MIL-STD-785, Tasks 205 and 206; and parts derating in accordance with the requirements of TE000-AB-GTP-010.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Qualification inspection.

INSPECTION	INVERTER SAMPLE				REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
	1	2	3	4		
Visual & mechanical inspection	X	X	X	X	3.4, 3.5, 3.6, 3.9, 3.10	4.7.2.1
Voltage and frequency regulation					3.7.1.2, 3.7.1.3, 3.7.2.1	4.7.2.2
Room ambient temp-alt	X*	X*	X	X		4.7.2.2.1
Power factors	X*	X*				4.7.2.2.2
Maximum temp-alt	X*	X*				4.7.2.2.3
Minimum temp-alt	X*	X*				4.7.2.2.4
Operating position	X	X				4.7.2.2.5
Worst condition	X	X				4.7.2.2.6
Efficiency	X*	X*	X	X	3.7.1.4	4.7.2.3
Three-phase output	X*	X*	X	X	3.7.1.5	4.7.2.4
Recovery time	X	X	X	X	3.7.1.6	4.7.2.5
Input power application		X		X	3.7.1.7	4.7.2.6
Overload & short circuit capacity	X*	X*	X	X	3.7.1.8	4.7.2.7
Electromagnetic compatibility	X		X		3.7.1.9	4.7.2.8
Output modulation	X*	X*	X	X	3.7.1.10	4.7.2.9
Waveform	X*	X*	X	X	3.7.1.11	4.7.2.10
Reduced input voltage operation	X	X			3.7.1.12	4.7.2.11
Acoustical noise	X				3.7.1.13	4.7.2.12
Overspeed (rotary inverters only)			X		3.7.1.14	4.7.2.13
Acceleration	X				3.7.2.2	4.7.2.14
Humidity	X				3.7.2.3	4.7.2.15
Shock	X				3.7.2.4	4.7.2.16
Vibration		X	X		3.7.2.5	4.7.2.17
Fungus				X	3.7.2.6	4.7.2.18
Salt fog		X			3.7.2.7	4.7.2.19
Dust				X	3.7.2.8	4.7.2.20
Explosive atmosphere (when specified)			X		3.7.2.9	4.7.2.21
Reliability development			X	X	3.8	4.7.3
Disassembly & inspection			X	X	---	4.7.4

Note: For three-phase units, all tests shall be conducted in the three-phase wye configuration. For units designed to operate in both wye and delta configuration, tests indicated by * shall be conducted in both delta and wye configurations.

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TABLE II. Quality conformance inspection.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
Visual & mechanical inspection	3.4, 3.5, 3.6, 3.9, 3.10	4.7.2.1
Voltage and frequency regulation at room ambient temperature-altitude	3.7.1.2, 3.7.1.3	4.7.2.2.1
Power factors	3.7.1.2, 3.7.1.3	4.7.2.2.2
Efficiency	3.7.1.4	4.7.2.3
Phase rotation (three-phase units only)	3.7.1.5.1	4.7.2.4.2
Phase balance (three-phase units only)	3.7.1.5.2	4.7.2.4.1
Output voltage modulation	3.7.1.10.1	4.7.2.9.1

Note: For three-phase units, all tests shall be conducted in the three-phase wye configuration.

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TABLE III. Maximum Temperature-Altitude Conditions.

CLASS A INVERTERS		CLASS B INVERTERS		CLASS C INVERTERS	
AMBIENT TEMPERATURE (°C)	ALTITUDE (FEET)	AMBIENT TEMPERATURE (°C)	ALTITUDE (FEET)	AMBIENT TEMPERATURE (°C)	ALTITUDE (FEET)
+71	Sea level	+85	Sea level	120	Sea level
+25	15,000	+85	17,500	80	25,000
+8	25,000	+50	35,000	60	37,500
-10	35,000	+20	50,000	40	50,000
-10	50,000	+20	65,000*	-12	65,000*

*Class B power rating lowered 50 percent above 50,000 feet.

TABLE IV. Minimum Temperature-Altitude Conditions.

CLASS A, B, AND C INVERTERS	
AMBIENT TEMPERATURE (°C)	ALTITUDE (FEET)
-55	Sea level
-55	10,000
-55	25,000
-55	50,000
-55	65,000*

*Class B power rating lowered 50 percent above 50,000 feet.

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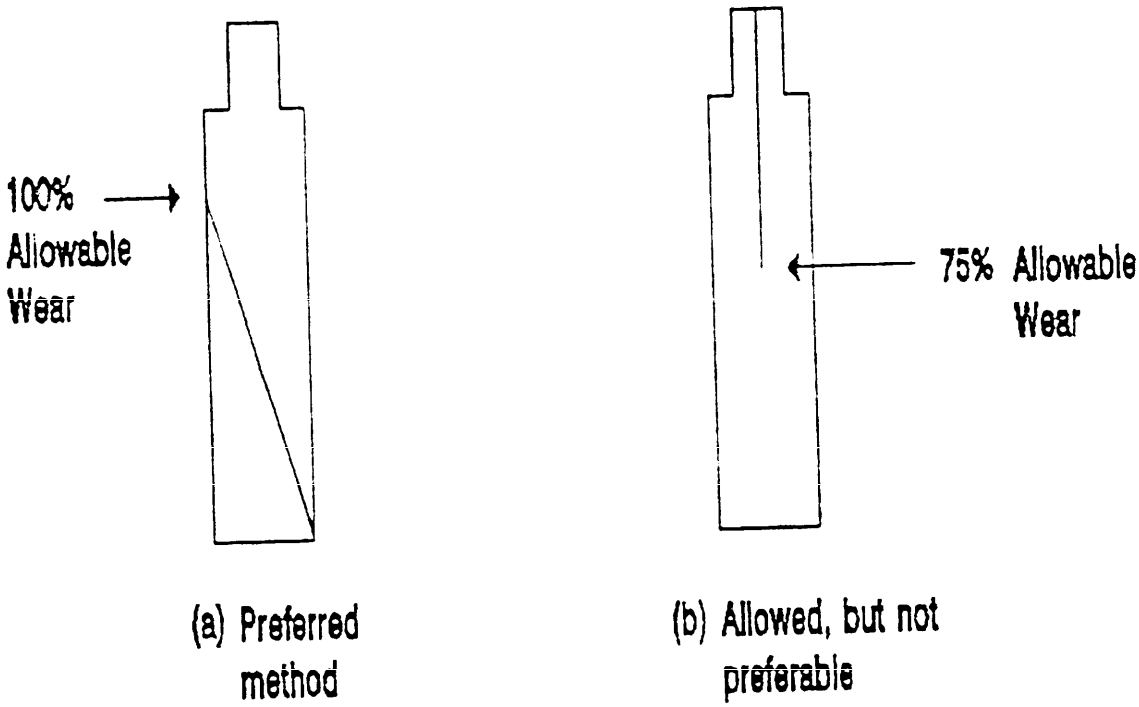


Figure '. Brush markings.

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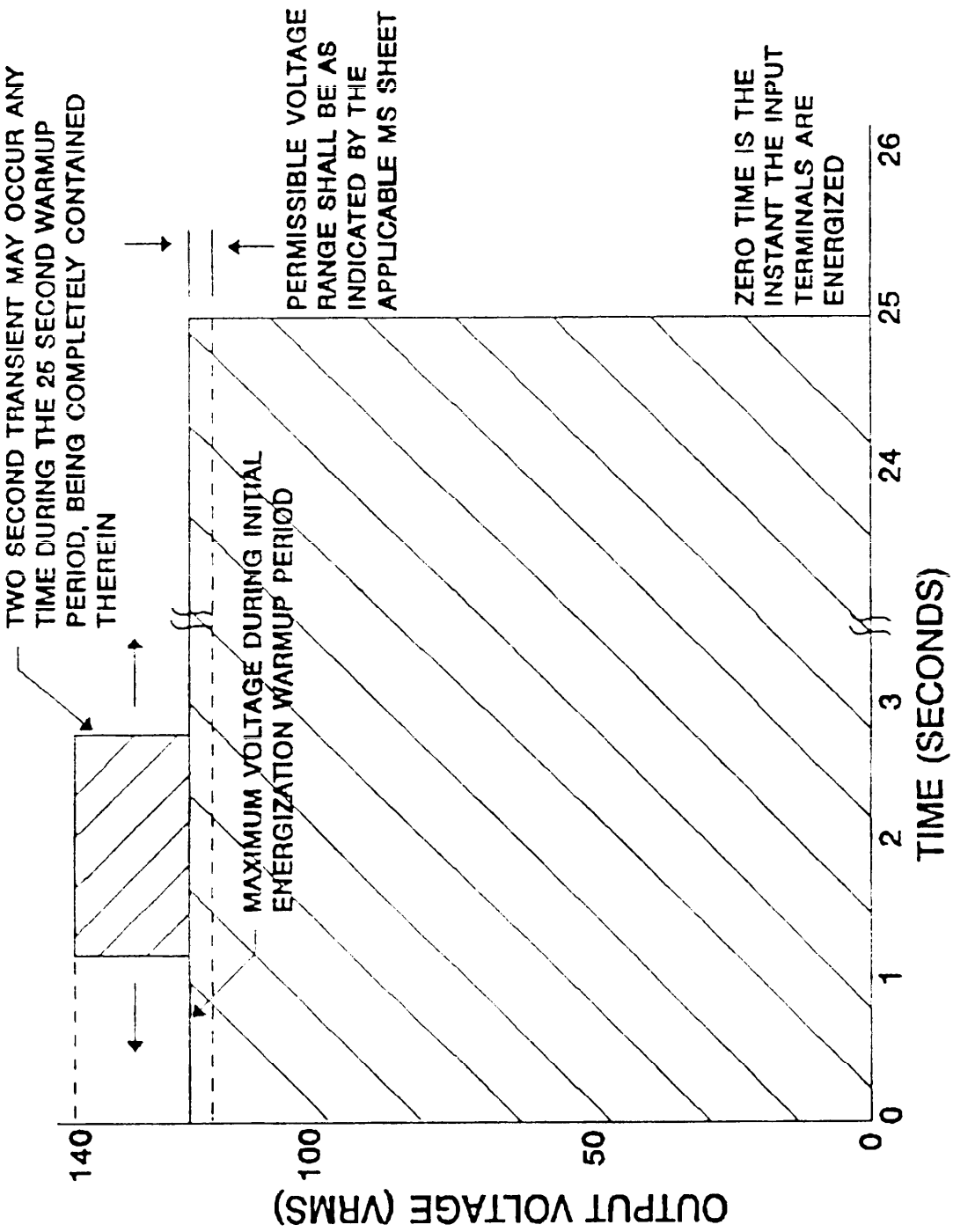


Figure 2. Output voltage ring input power application.

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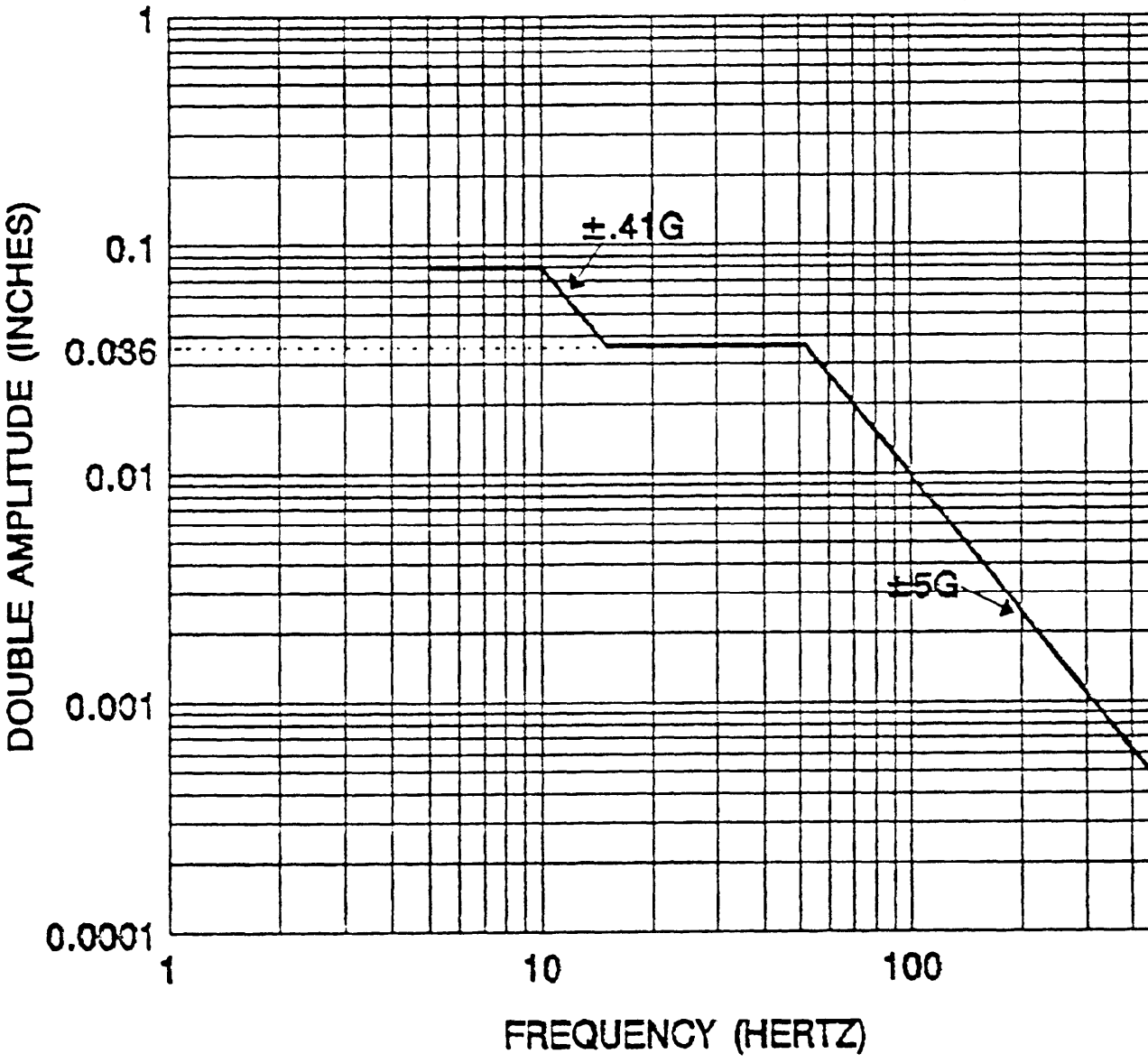


Figure 3. Vibration test curve.

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review Activities:
Army - ME
Air Force - 80

Preparing Activity
Navy - AS
Project No. 6125-A33

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-7032G	2. DOCUMENT DATE (YYMMDD) 8 August 1994

3. DOCUMENT TITLE
Inverters, Aircraft, General Specification for

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Electrical Power Section (AIR-53624)	b. TELEPHONE (Include Area Code) (1) Commercial: (703) 604-3290 (2) AUTOVON: 664-3290
c. ADDRESS (Include Zip Code) Naval Air Systems Command 1421 Jefferson Davis Hwy Arlington, VA 22243-5360	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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